

IZUHARA, Japan

WMO: 478000

Lat: 34.1978N

Lon: 129.2919E

Elev: 4

StdP: 101.28

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.0	0.1	-12.1	1.3	1.7	-10.5	1.5	2.4	9.9	4.3	9.1	3.6	3.5	330	0.391

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.1	31.8	26.3	30.8	26.0	29.8	25.6	27.0	30.4	26.5	29.7	26.1	29.0	3.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.5	29.0	25.6	20.8	28.7	25.1	20.2	28.3	85.2	30.7	82.9	29.7	80.9	29.1	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.3	7.1	-2.8	33.4	1.3	1.4	-3.8	34.4	-4.6	35.1	-5.4	35.9	-6.3	36.9		
(5)			-5.4	27.9	1.3	0.8	-6.3	28.5	-7.1	29.0	-7.8	29.4	-8.7	30.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	5.7	6.8	9.9	14.2	18.3	21.4	25.8	27.2	23.6	18.9	13.3	7.9	(6)
(7)		DBStd	7.71	3.06	3.16	3.04	2.62	2.20	2.03	2.51	1.95	2.05	2.59	3.40	3.38	(7)
(8)		HDD10.0	369	138	97	38	2	0	0	0	0	0	0	10	83	(8)
(9)		HDD18.3	1632	393	323	261	126	28	1	0	0	0	23	153	323	(9)
(10)		CDD10.0	2607	4	8	36	128	257	343	489	532	407	278	109	18	(10)
(11)		CDD18.3	828	0	0	0	2	27	94	231	273	157	42	2	0	(11)
(12)		CDH23.3	5962	0	0	0	2	68	293	1971	2764	795	69	1	0	(12)
(13)		CDH26.7	1568	0	0	0	0	4	21	528	897	116	2	0	0	(13)
(14)	Wind	WSAvg	3.0	3.3	3.2	3.3	3.2	2.9	2.6	3.2	2.8	2.7	2.8	2.8	3.1	(14)
(15)	Precipitation	PrecAvg	2228	76	92	151	219	224	298	340	281	242	118	105	61	(15)
(16)		PrecMax	3486	231	242	353	542	502	1101	854	992	616	303	329	193	(16)
(17)		PrecMin	1433	0	6	15	44	16	32	33	15	32	5	0	2	(17)
(18)		PrecStd	465	48	51	73	103	111	177	206	182	142	76	71	44	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.7	17.4	19.4	23.2	27.1	28.7	32.8	33.6	30.6	26.3	22.5	17.7	(19)
(20)			MCWB	13.1	14.7	14.1	15.4	17.0	21.9	26.3	26.8	25.7	21.7	18.4	14.4	(20)
(21)		2%	DB	13.5	15.3	17.7	21.4	25.1	27.1	31.5	32.2	29.3	25.1	21.0	15.9	(21)
(22)			MCWB	10.3	12.3	13.4	15.6	17.3	22.1	26.0	26.5	25.0	21.1	17.2	12.3	(22)
(23)		5%	DB	11.9	13.4	16.4	20.1	23.8	26.0	30.5	31.2	28.0	24.0	19.7	14.6	(23)
(24)			MCWB	8.6	10.0	12.5	15.2	17.5	22.1	25.7	26.2	24.2	19.7	16.1	11.0	(24)
(25)		10%	DB	10.6	11.9	15.1	18.9	22.7	25.0	29.4	30.4	26.9	23.0	18.6	13.5	(25)
(26)			MCWB	7.5	8.4	11.3	14.5	17.2	21.9	25.3	26.0	23.3	18.7	14.9	10.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	15.4	16.2	19.6	22.2	24.9	27.4	27.9	26.7	23.6	19.9	15.9	(27)
(28)			MCDWB	15.0	16.9	17.6	20.8	23.3	26.6	31.1	31.5	29.4	25.4	21.6	17.0	(28)
(29)		2%	WB	11.0	13.1	14.7	17.7	20.4	24.2	26.6	27.2	25.7	22.3	18.3	13.2	(29)
(30)			MCDWB	12.7	14.6	16.6	19.6	22.4	25.7	30.1	30.6	28.2	24.4	20.1	15.2	(30)
(31)		5%	WB	9.2	10.8	13.3	16.4	19.4	23.4	26.1	26.8	25.0	20.7	16.9	11.7	(31)
(32)			MCDWB	11.3	12.7	15.7	18.9	21.9	25.1	29.3	30.1	27.2	22.9	19.0	13.9	(32)
(33)		10%	WB	7.8	9.0	11.9	15.2	18.5	22.4	25.7	26.4	24.1	19.6	15.5	10.4	(33)
(34)			MCDWB	10.2	11.1	14.5	17.7	21.3	24.4	28.6	29.5	26.1	22.2	18.0	13.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.0	6.5	6.6	6.9	7.0	5.4	4.6	5.1	5.1	6.2	6.8	6.5	(35)
(36)			MCDBR	7.5	8.3	8.2	8.9	9.2	7.2	6.2	6.5	6.5	6.9	7.5	7.7	(36)
(37)			MCWBR	5.9	6.5	6.1	5.9	5.5	4.1	2.9	3.0	3.5	4.1	5.2	5.7	(37)
(38)		5% WB	MCDWR	6.7	7.2	7.4	6.7	7.1	4.9	5.5	5.8	5.4	5.5	6.4	6.9	(38)
(39)			MCWBR	6.2	6.8	6.5	5.7	5.3	3.5	2.8	2.9	3.3	4.0	5.3	6.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.441	0.490	0.558	0.581	0.589	0.614	0.592	0.549	0.513	0.476	0.444	0.430	(40)	
(41)		taud	2.024	1.931	1.783	1.781	1.817	1.848	1.951	2.078	2.109	2.087	2.097	2.060	(41)	
(42)		Ebn at Noon	729	737	721	730	730	711	724	748	755	744	723	711	(42)	
(43)		Edh at Noon	136	167	209	219	214	207	186	161	149	141	126	125	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.39	3.08	4.14	4.91	5.34	4.65	4.94	5.07	4.07	3.55	2.65	2.22	(44)	
(45)		RadStd	0.20	0.35	0.34	0.39	0.63	0.46	0.77	0.57	0.40	0.35	0.29	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page